



**PES University, Bengaluru**  
(Established under Karnataka Act 16 of 2013)

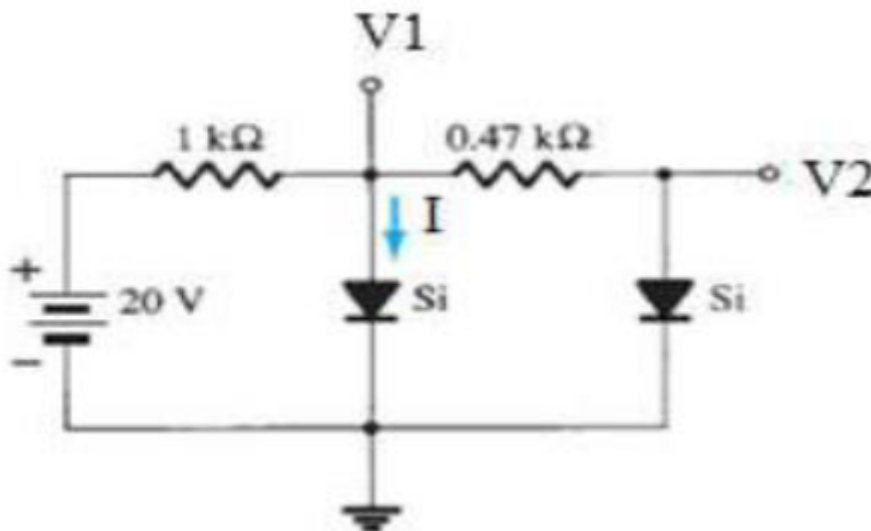
**END SEMESTER ASSESSMENT (ESA) - JAN 2023**

**UE23EC141A - Electronic Principles And Devices**

**Total Marks : 100.0**

1.a. Describe the Temperature Effects on V-I Characteristics(Forward and Reverse) of a Diode with suitable diagrams? (2M + 4M) (6.0 Marks)

1.b. Determine Current (I), Voltages ( $V_1$ ) and ( $V_2$ ) using second approximation for a diode in circuit diagram (1M+1M+ 1M +1M) (4.0 Marks)



1.c. The reverse saturation current of a Germanium diode is  $200\mu\text{A}$  at room temperature of  $27^\circ\text{C}$ . Calculate the current in forward biased condition, if forward biased voltage is  $0.2\text{V}$  at room temperature. If temperature is increased by  $30^\circ\text{C}$ , calculate the reverse saturation current and the forward current for the same forward voltage at new temperature. Given  $K = 1.38 \times 10^{-23} \text{ Joule/Kelvin}$ ,  $q = 1.60 \times 10^{-19} \text{ coulomb}$ . (each step 1 mark) (6.0 Marks)

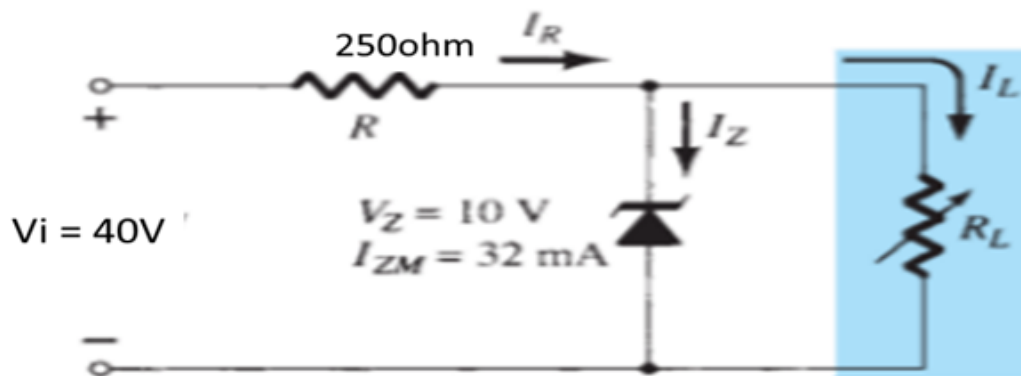
1.d. (i) With a neat diagram, explain the V-I characteristics of NPN BJT Common emitter configuration. And find the value of  $I_C$ ,  $\alpha$  and  $\beta$  if  $I_E = 1.2\text{mA}$  and  $I_B = 30\mu\text{A}$ . (2M+2M+2M = 6 Marks)  
(ii) Draw n-channel E-Type MOSFET ( 3 M) (9.0 Marks)

2.a. With a neat circuit diagram and Waveform explain the working principle of Center Tapped Full wave Rectifier. (2M+ 1M+3M) (6.0 Marks)

2.b. Derive the expression for  $V_{dc}$ ,  $V_{rms}$  for Half wave Rectifier and hence find the ripple factor of the Rectifier. (3M+ 3M+3M) (9.0 Marks)

2.c. A bridge Full wave rectifier using silicon diodes has a load resistance of 1.5K ohm. The rms voltage across the secondary is 20V. Calculate Peak value of load current, Average load current and rms output voltage. (4M) And also for same Full wave rectifier calculate the dc voltage if capacitor filter with 500 $\mu$ F capacitor and a load current of 200mA with 8% ripple. (2M) (6.0 Marks)

2.d. Find the range of load resistance and the maximum power rating for the circuit? (1M +2M +1 M)



(4.0 Marks)

3.a. Realize the following Gates using Universal Gate (2M+2M+1M+2M)

(i) XNOR gate using NAND gate(2M) (ii) AND gate using NOR gate (2M)

(iii)NOT gate using NOR gate(1M) (iv) OR Gate using NAND gate(2M)

(7.0 Marks)

3.b. Simplify the given Boolean expression and implement the simplified expression using Basic gates.  **$f(x,y,z) = x'yz + xyz' + xyz + xy'z$**  (3M+ 3M)

(6.0 Marks)

3.c. Describe 4: 1 Multiplexer with Block diagram, Function Table, Boolean Function and Logic diagram using Basic Gates (1M+ 1M+1M+2 M)

(6.0 Marks)

3.d. Explain J-K Flip flop with Circuit diagram and Characteristic Table (1M + 2M + 3M) (6.0 Marks)

4.a. Define Roaming in Cellular Technology and describe the issues involved in Roaming (2M+ 6M) (8.0 Marks)

4.b. Describe the I2C interface with neat diagram (2M+3M) (5.0 Marks)

4.c. Discuss the differences between (any 2 differences)

(i) Flash and SRAM (2M)

(ii) Microcontroller and Microprocessor (2M)

(4.0 Marks)

4.d. Draw the Data Flow Model of ARM Processor and explain the same.

(4M+4M)

(8.0 Marks)



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**UE23EC141A - Electronic Principles And Devices**

**Total Marks : 100.0**

1.a. Explain the following:

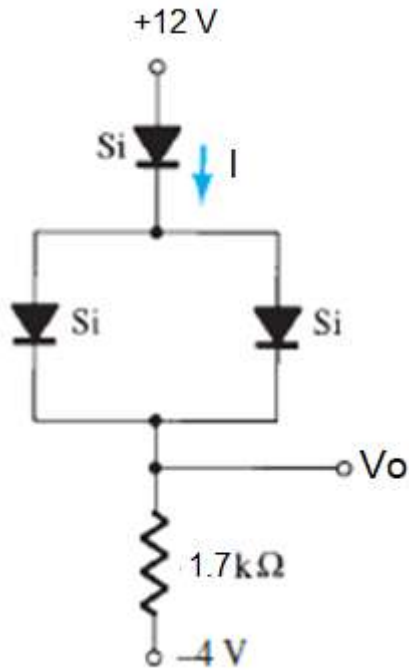
- (i) First and Second Diode Approximations with necessary circuit diagrams and waveforms.
- (ii) Illustration of logical OR operation using Diodes with circuit diagram and Truth table.
- (iii) Avalanche breakdown

**(8.0 Marks)**

1.b. Using Shockley's equation, find the diode current  $I_D$  for a Silicon diode, if the applied voltage  $V_D = 0.73$  volts and reverse saturation current is  $2 \times 10^{-12}$  A at a temp of  $30^\circ$  C. Consider ( $\eta=1$ ). Find the new current  $I_D$ , if the Temperature is increased to  $50^\circ$  C. **(5.0 Marks)**

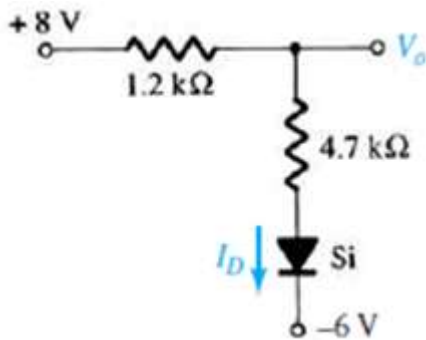
1.c. Use Second approximation for a diode to determine the unknown values for the following:

(i) Determine  $I$  and  $V_o$  for the circuit shown in the figure below:



(ii) Determine  $I_D$  and  $V_o$  for the circuit shown in the figure below:

(4.0 Marks)



1.d. With a neat diagram, explain the Input and output V-I characteristics of a PNP Common Base BJT.

Find the value of  $\alpha$  and  $I_B$ , if  $I_E = 1.32\text{mA}$  and  $I_C = 1.11\text{mA}$ . (8.0 Marks)

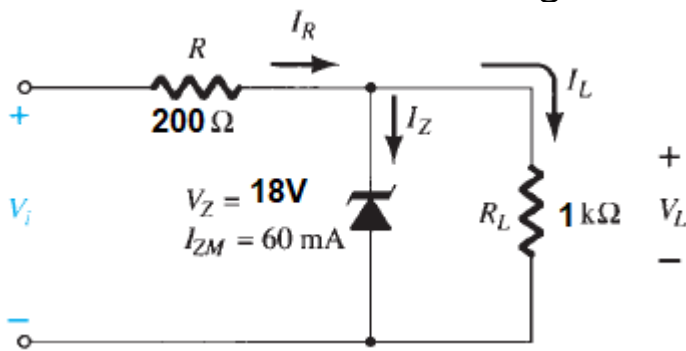
2.a. With a neat diagram, explain the working principle of a Full wave Rectifier with Centre tapped Transformer and derive the expression for  $V_{dc}$  and  $V_{rms}$  and hence find the efficiency of the Rectifier. (10.0 Marks)

2.b. With a neat diagram and waveform, explain the working principle of a Full wave Bridge Rectifier with Capacitive Filter. (6.0 Marks)

2.c. For a half wave rectifier with ideal diode, an ac voltage of RMS value 20V, 50Hz is applied. If the load resistance is  $200\Omega$ , find the values of  $V_m$ , dc output voltage, output frequency and ripple factor across the load resistance. (4.0 Marks)

2.d. Determine the range of values of  $V_i$  that will maintain the Zener diode in the "on" state for the following Circuit:

(5.0 Marks)



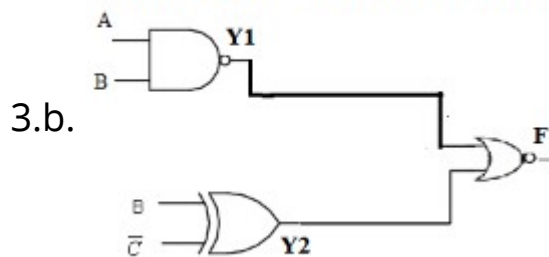
3.a. Realize The following Logic gates using NAND Gates only:

(i) NOR (ii) XOR (iii) AND (iv) NOT

(6.0 Marks)

3.b. Write the expressions for  $Y_1$ ,  $Y_2$  and the simplified expression for  $F$

(5.0 Marks)



3.c. Draw the circuit diagram and truth table for the following flip-flops using NAND gates:

(i) S-R Flip Flop

(ii) J-K Flip Flop

(8.0 Marks)

3.d. Define Asynchronous Counters. Design a 3-bit Asynchronous Mod-8 up-counter using positive edge triggered T Flip-flops and write its truth table. (6.0 Marks)

4.a. Explain Communication System with a neat block diagram (6.0 Marks)

4.b. With respect to Cellular Communication, define the following:

- 1) Hand off
- 2) Roaming
- 3) Co-channel Interference

(6.0 Marks)

4.c. Explain the Data Flow Model of ARM Processor with a neat diagram.

(8.0 Marks)

4.d. Briefly explain the different operating modes of ARM Processor

(5.0 Marks)



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**END SEMESTER ASSESSMENT (ESA) - Jan-Feb**

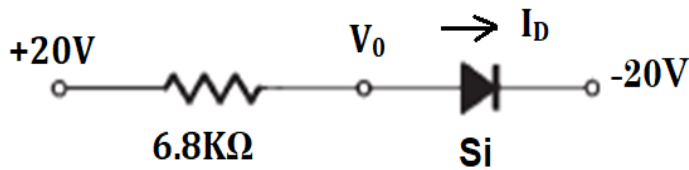
**UE24EC141A - Electronic Principles And Devices**

**Total Marks : 100.0**

1.a. (i) Determine the dynamic resistance of a diode at a temperature of  $40^{\circ}\text{C}$  and a forward bias voltage of  $0.6\text{V}$ , if the reverse saturation current is  $5\text{pA}$  at  $300\text{K}$ .  $n = 1$  **(4M)**

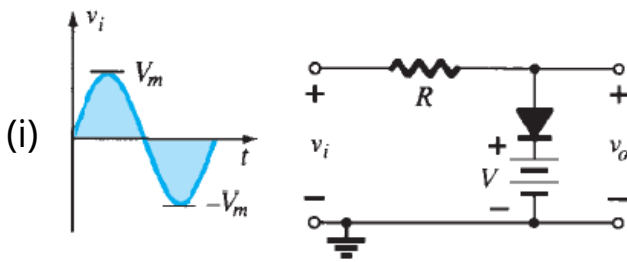
(ii) Determine  $I_D$  and  $V_0$  in the circuit shown below **(4M)**

**(8.0 Marks)**

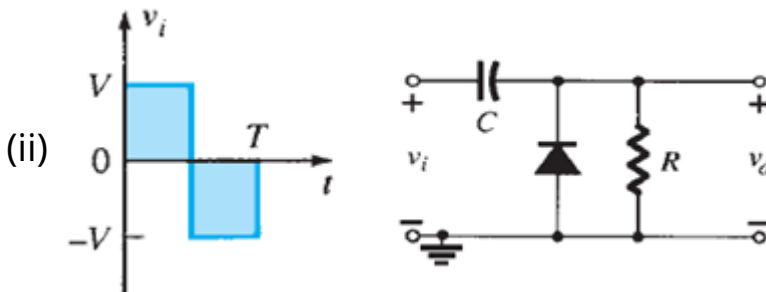


1.b. What are the different approximations used to represent the diode? For each of them draw the equivalent circuit and VI Characteristics **(9.0 Marks)**

1.c. Briefly explain the working of the following circuits and draw their output waveforms assuming ideal diodes **(4M each)**

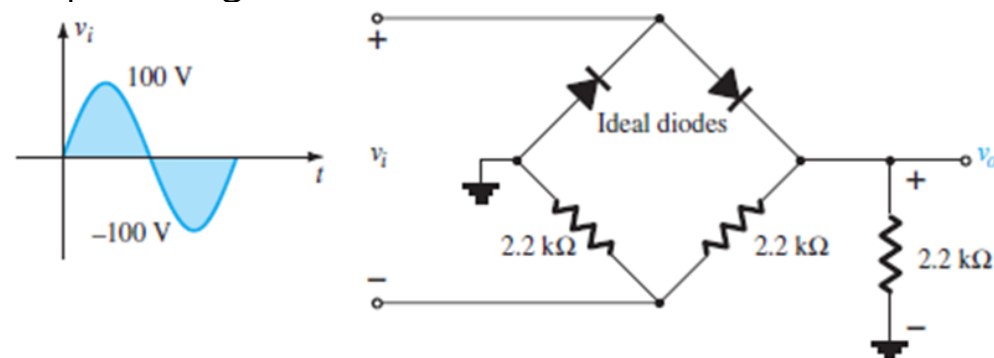


(8.0 Marks)



2.a. (i) With the circuit explain how PIV of the diode is determined in a bridge rectifier. **(3M)**

(ii) Draw the output waveform and determine the average value of the output voltage in the circuit shown below. **(5M)**



(8.0 Marks)

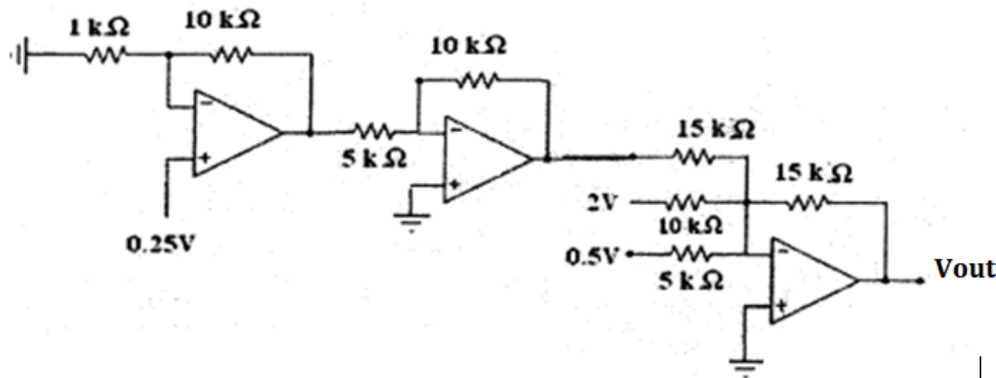
2.b. Draw the circuit of a full-wave rectifier with C filter and explain its working. Draw the output waveform. If  $R_L = 500\Omega$ ,  $C = 500\mu\text{F}$  and average load current is 20mA for the same circuit, determine the RMS value of the ripple voltage. Assume  $f = 50\text{Hz}$  **(9.0 Marks)**

2.c. (i) What are the conditions for a zener diode to operate as a voltage regulator? **(2M)**

(ii) Determine  $R_{Lmin}$  and  $R_{Lmax}$  to obtain an output voltage of 10V for a zener regulator with  $V_i = 20V$ ,  $P_{Zmax} = 400 \text{ mW}$ ,  $I_{zmin} = 1 \text{ mA}$  and  $R = 220\Omega$ . **(6M)** (8.0 Marks)

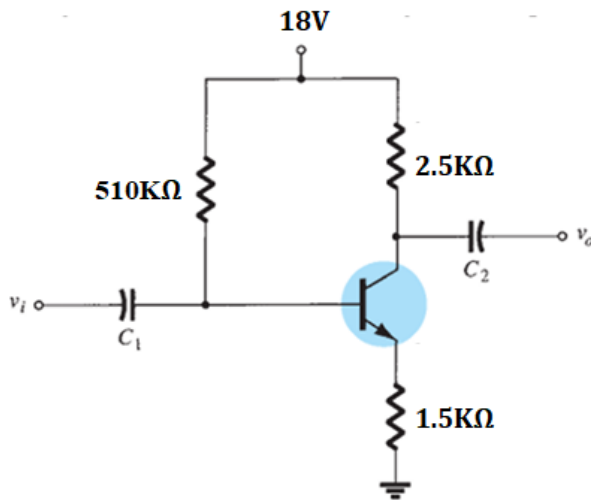
3.a. Draw the output characteristics of a transistor in CE configuration and mark the different regions of operation. Write the biasing condition in each of the regions. Write the current equation in the active region **(6.0 Marks)**

3.b. Identify each stage of the op-amp and determine  $V_{out}$  for the circuit shown. Write the relevant equations **(7.0 Marks)**



3.c. Determine  $I_C$ ,  $V_{CE}$ ,  $I_{C(sat)}$  and  $P_C$  for the circuit shown below. Comment on the location of this operating point. Given  $\beta = 100$

(7.0 Marks)



3.d. With a circuit diagram, derive an expression for the output voltage of op-amp integrator. (5.0 Marks)

4.a. Draw the circuit of a Full - Adder using XOR and basic gates and write its Truth Table. Write the expressions for its outputs. (6.0 Marks)

4.b. (i) Convert the decimal number  $385.625_{10}$  to binary and hexadecimal numbers. Write the steps.

(ii) Perform subtraction using 2's complement method

$$37_{10} - 23_{10}$$

(6.0 Marks)

4.c. Simplify the following Boolean expression and realise the simplified expression using NAND gates.

$$Y = (A + \overline{B} + \overline{C}) (\overline{A} + \overline{B} + \overline{C}) (\overline{A} + B)$$

(5.0 Marks)

4.d. Draw the circuit of a Clocked- D Flip-Flop using NAND gates and write its truth table. Draw a 4-bit Serial-in-serial-out shift register using positive edge-triggered D Flip-Flops. A data of 1011 is to be shifted into the register starting from its LSB. Write the state table showing how data is shifted into it. Assume the initial contents of the register to be 0000.

(8.0 Marks)



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**END SEMESTER ASSESSMENT (ESA) - Jan-Feb**

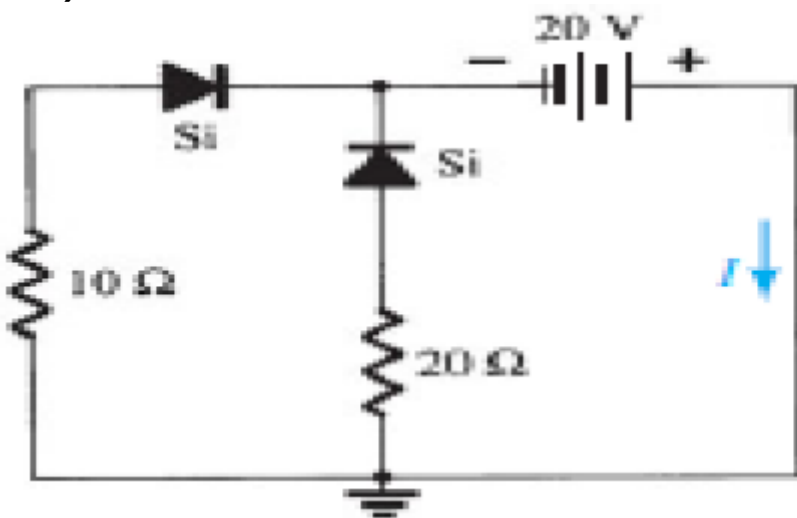
**UE24EC141A - Electronic Principles And Devices**

**Total Marks : 100.0**

**1.a. Describe the Temperature Effects on V-I Characteristics of a Diode with Forward and Reverse biased and suitable diagrams?**  
(Diagram-2 M + Forward biased - 2 M + Reverse biased -2 M) (6.0 Marks)

**1.b. Determine the voltages and current through  $20\Omega$  and  $10\Omega$  resistor and also the total Current (I) for the configuration ( 2M + 1M +1M)**

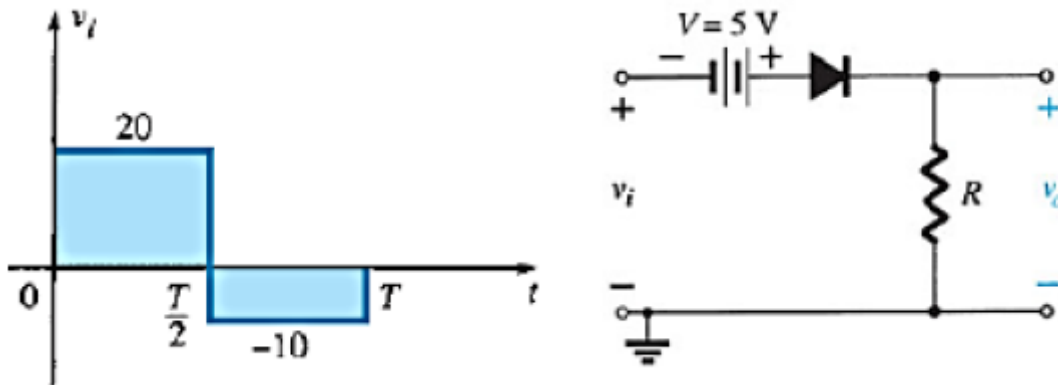
(4.0 Marks)



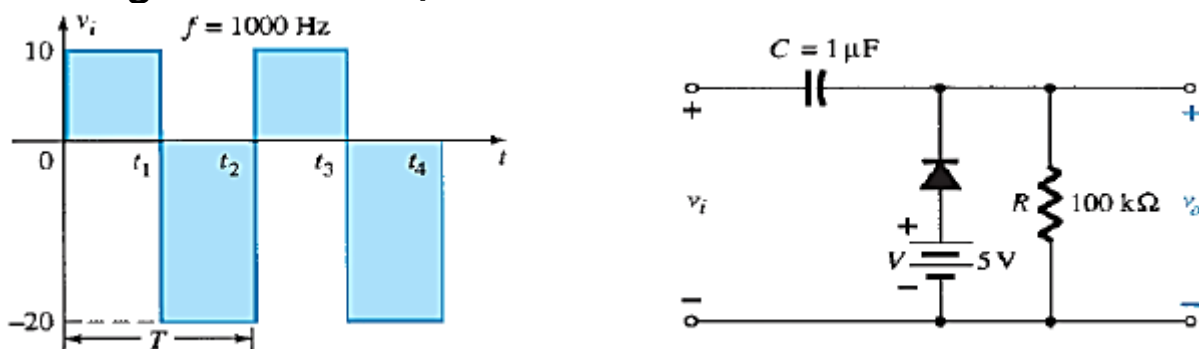
1.c. The reverse saturation current of a Germanium diode is  $200\mu\text{A}$  at room temperature of  $27^\circ\text{C}$ . Calculate the current in forward biased condition, if forward biased voltage is  $0.2\text{V}$  at room temperature. If temperature is increased by  $30^\circ\text{C}$ , calculate the reverse saturation current and the forward current for the same forward voltage at new temperature.

Given:  $K=1.38 \times 10^{-23} \text{Joule/Kelvin}$ ,  $q=1.60 \times 10^{-19} \text{ coulomb}$  (each step 1 mark). (6.0 Marks)

1.d. (i) Find the output voltage for the Clipper network with Ideal diode, if the applied signal is the square wave of Fig. ( Calculation : 2marks and Diagrams : 2marks)



(ii) Determine  $v_o$  for the network of shown below with Ideal diode, for the input indicated in Fig below - 5 marks ( Calculation : 3marks and Diagrams : 2marks)



(9.0 Marks)

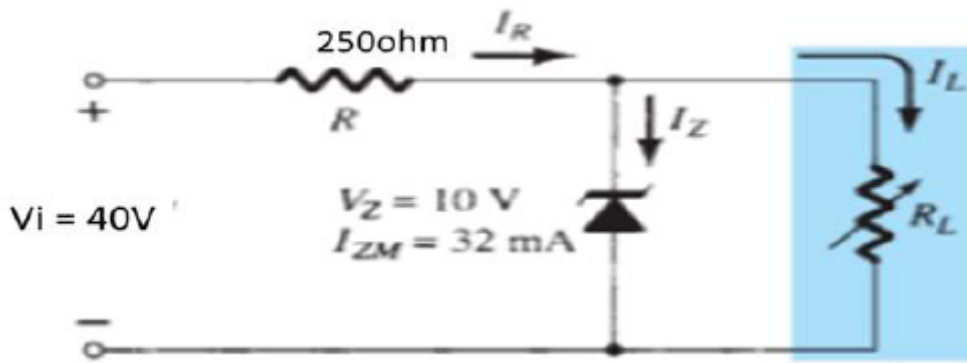
**2.a. With a neat circuit diagram and waveform explain the working principle of Center Tapped Full wave Rectifier ? (Circuit diagram- 2marks, Waveform- 1 mark and Explanation- 3 marks) (6.0 Marks)**

**2.b. Derive the expression for  $V_{dc}$ ,  $V_{rms}$  for Half wave Rectifier (Ideal Diode) and hence find the ripple factor of the Rectifier. (Solve for  $V_{dc}$ - 3marks ,  $V_{rms}$  - 3marks and Ripple factor- 2marks) (8.0 Marks)**

**2.c. A bridge Full wave rectifier using silicon diodes has a load resistance of 1.5K ohm. The rms voltage across the secondary is 20V. Calculate Peak value of load current, Average load current and rms output voltage. (4 Marks)  
And also for same Full wave rectifier calculate the dc voltage if capacitor filter with 500 $\mu$ F capacitor and a load current of 200mA with 8% ripple. (2Marks) (6.0 Marks)**

2.d. Find the range of load resistance and the maximum power rating for the circuit? ( $R_{Lmin} = 1M$ , Solve  $I_{Lmin}$  and  $R_{Lmax}$  - 3Marks and  $P_z$  - 1Mark)

(5.0 Marks)



3.a. (i) What is Base Width Modulation or Early Effect ? (2 Marks)

(ii) Derive relationship between alpha and Beta for the BJT? (2Marks)

(iii) In a common base connection, current amplification factor is 0.9.

If the emitter current is 1mA, determine the value of base current (2 Marks)

(6.0 Marks)

3.b. Determine the following for the fixed-bias configuration of Fig. 4.7

- (a)  $I_{BQ}$  and  $I_{CQ}$  (2 Marks)
- (b)  $V_{CEQ}$  (1 Mark)
- (c)  $V_B$  and  $V_C$  (2 Marks)
- (d)  $V_{BC}$  (1 Mark)

(6.0 Marks)

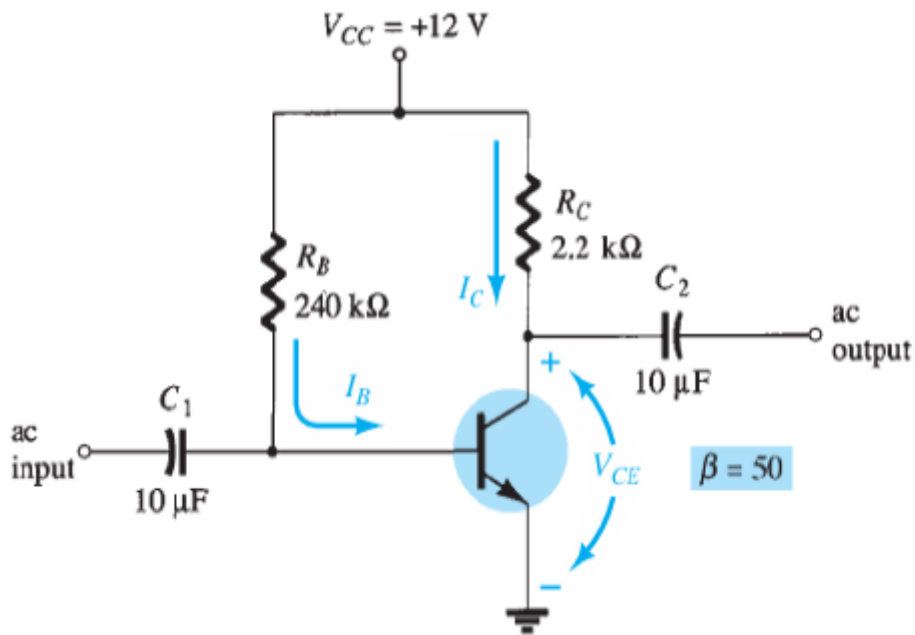
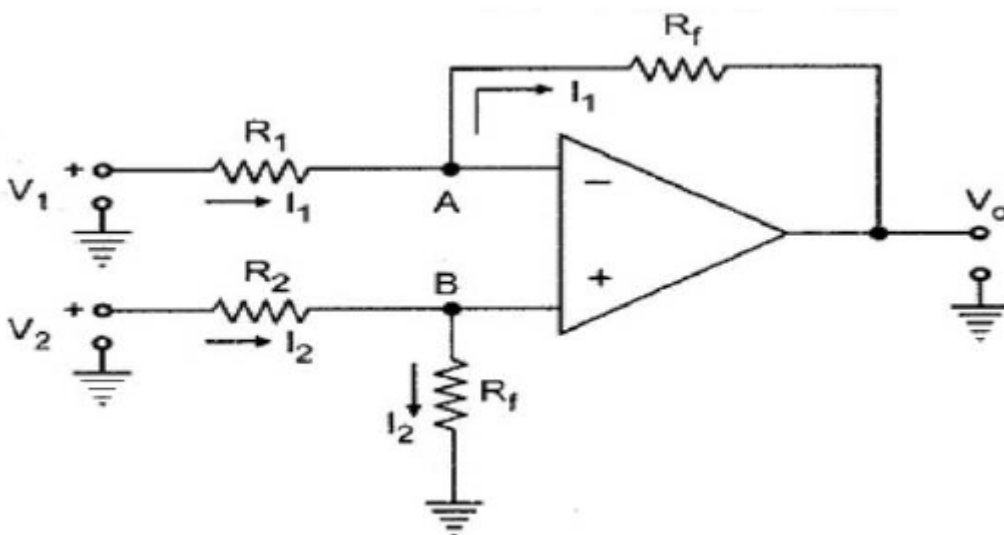


FIG. 4.7

3.c. Derive an Output Equation for the Voltage Subtractor shown in the Figure below:

(8.0 Marks)

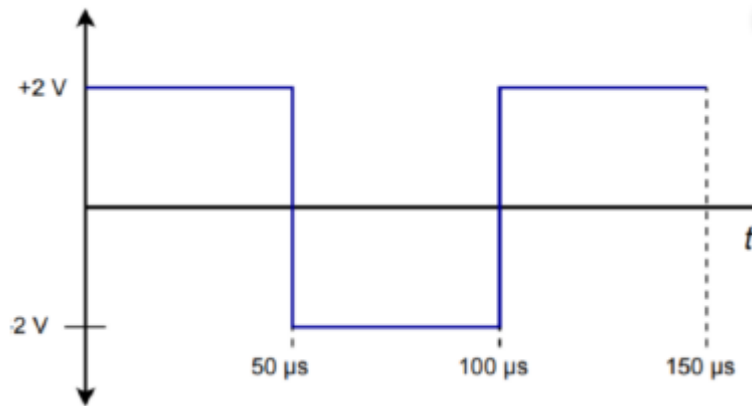


3.d. (i) Calculate the output voltage of an Inverting op-amp summing amplifier for the following set of voltages and resistors. Use  $R_f = 1 \text{ M ohm}$ .

$V_1 = +1 \text{ V}$ ,  $V_2 = +2 \text{ V}$ ,  $V_3 = +3 \text{ V}$ ,  $R_1 = 500 \text{ k ohm}$ ,  $R_2 = 1 \text{ M ohm}$ ,  $R_3 = 1 \text{ M ohm}$  - 2 Marks

(ii) Sketch the output of the integrator circuit if the input signal is a 10 kHz, 2 V peak square wave. Assume  $R=10\text{K}\Omega$  and  $C=10\text{nF}$ . (3 Marks)

(5.0 Marks)



(i) Subtract  $(1010)_2$  from  $(1000)_2$  using 2's complement.(2M)

4.a. (ii)Simplify the given Boolean expression and implement using NAND gates.

$$F = AB + A(B+C) + B(B+C) \text{ (3M + 2 M)}$$

(7.0 Marks)

4.b. Implement Full Adder using two Half Adder and one OR Gate?  
(Sum and Carry Equation – 2 Marks , Implementation – 4 Marks)

(6.0 Marks)

**4.c. Explain J-K Flip flop with NAND Gates, Write symbol, Circuit diagram and Characteristic Table (1M + 2M+ 3M) (6.0 Marks)**

**4.d. Explain 3 bit Asynchronous up- counter with Circuit diagram, Truth Table and Timing diagram? ( 2M + 1M + 3M ) (6.0 Marks)**



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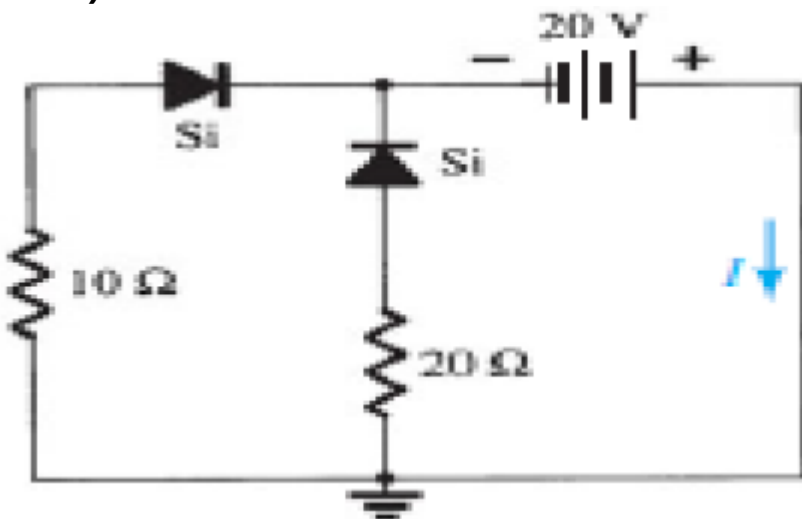
**END SEMESTER ASSESSMENT (ESA) - Feb 2026**

**UE25EC141A - Electronic Principles and Devices**

**Total Marks : 100.0**

**1.a. With neat diagram explain temperature effects on V-I Characteristics of a Diode when Forward bias and Reverse bias? (1M + 2M + 2M)** (5.0 Marks)

**1.b. Determine the voltages and current through  $20\Omega$  and  $10\Omega$  resistor and also the total Current (I) for the configuration ( 2M + 2M + 1 M)** (5.0 Marks)



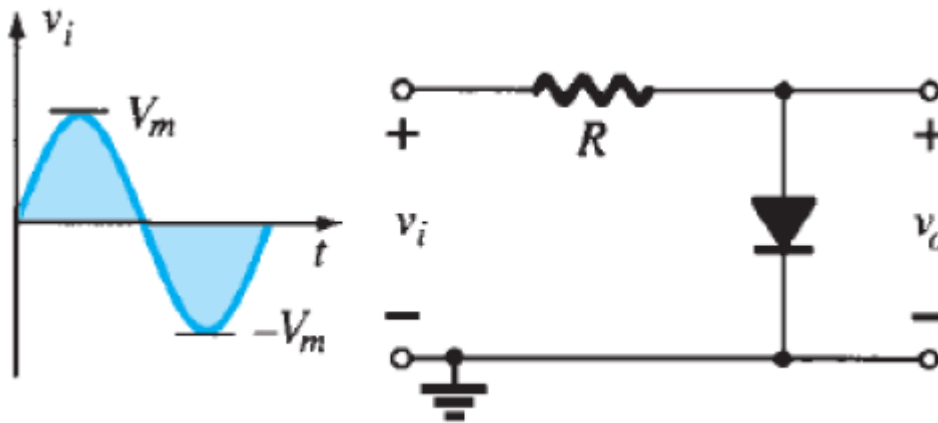
1.c. The reverse saturation current of a Germanium diode is  $200\mu\text{A}$  at room temperature of  $27^\circ\text{C}$ . Calculate the current in forward biased condition, if forward biased voltage is  $0.2\text{V}$  at room temperature.

If temperature is increased by  $30^\circ\text{C}$ , calculate the reverse saturation current and the forward current for the same forward voltage at new temperature.

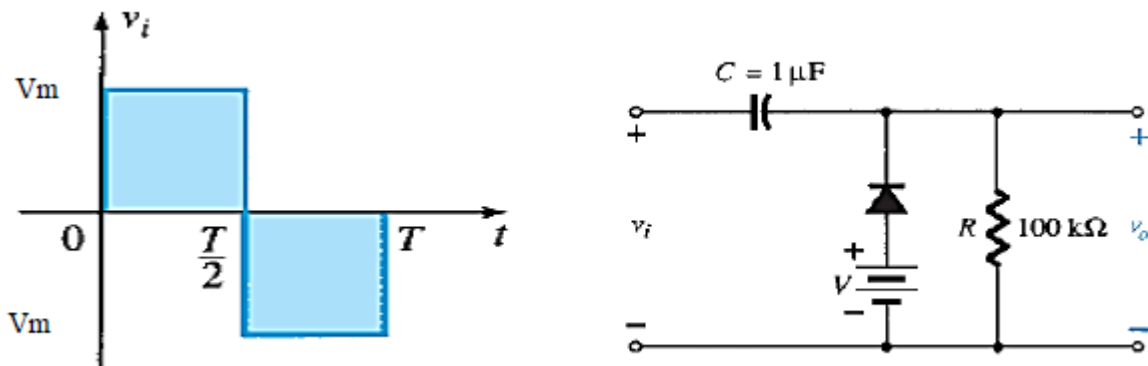
Given:  $K=1.38 \times 10^{-23} \text{Joule/Kelvin}$ ,  $q=1.60 \times 10^{-19} \text{coulomb}$  (each step 1 mark). (5.0 Marks)

1.d. Determine Output waveform  $V_o$  for the clipper network shown below with Ideal diode if sinusoidal input of peak  $V_m = 5\text{V}$  is applied. (2 M + 2M)

(4.0 Marks)



1.e. For the circuit below draw the output waveform during positive and negative half of input waveform with peak  $V_i = V_m$  for Ideal diode as shown. (3M + 3M)



(6.0 Marks)

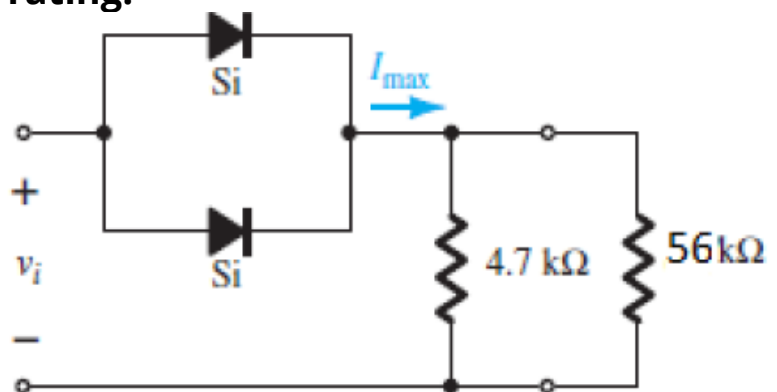
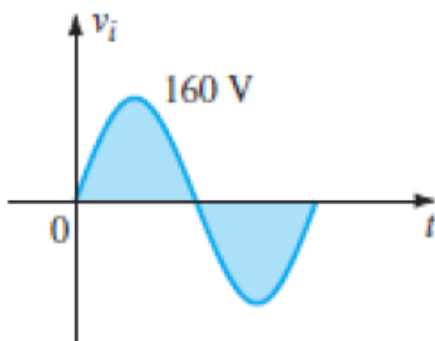
2.a. With a neat circuit diagram and waveform explain the working principle of Center Tapped Full wave Rectifier for Ideal Diode. (2M + 1M +3 M) (6.0 Marks)

2.b. (a) Given  $P_{\max} = 14 \text{ mW}$  for each diode shown, determine the maximum current rating of each diode (using the approximate equivalent model). (Each 1 M)

(b) Determine  $I_{\max}$  for  $V_{\max} = 160 \text{ V}$ .

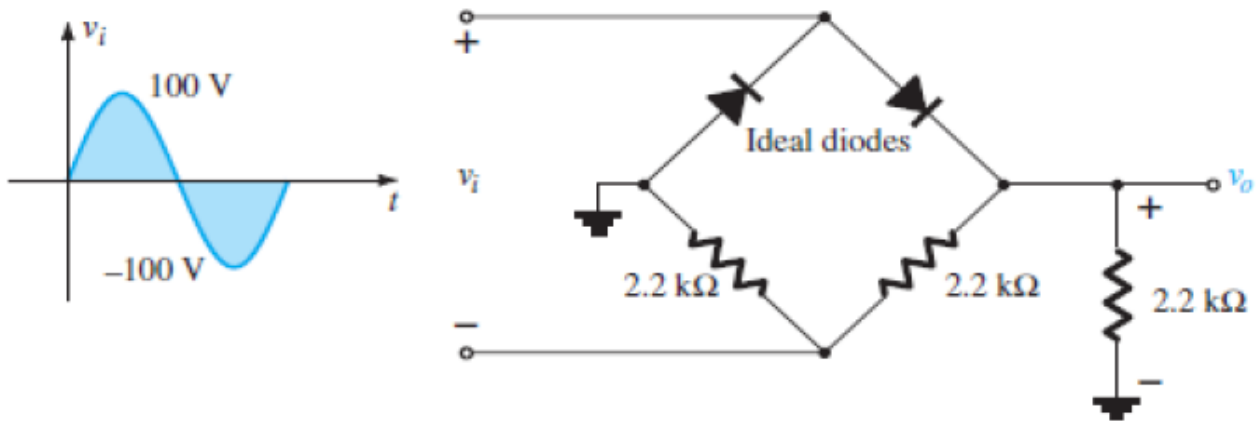
(c) Determine the current through each diode at  $V_{\max}$  using the results of part (b).

(d) If only one diode were present, determine the diode current and compare it to the maximum rating.



(4.0 Marks)

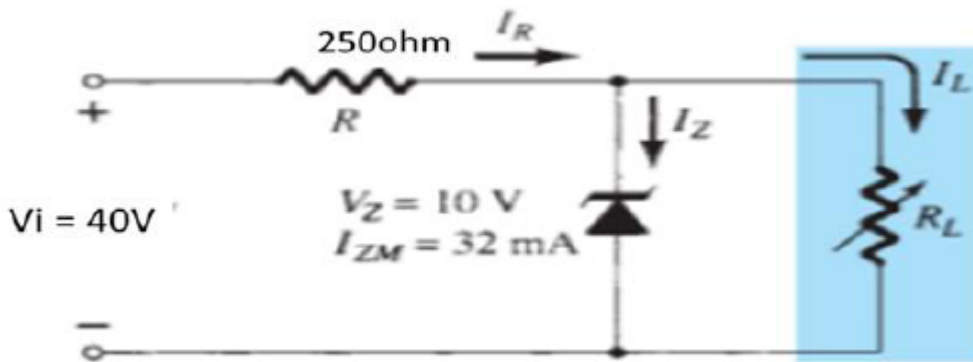
2.c. (a) Sketch  $V_o$  for the network of Fig below and determine the  $V_m$  voltage during Positive and Negative half cycle of the input waveform and  $V_{dc}$ . (4 Marks)



(b) Calculate the dc voltage if capacitor filter with  $500\mu\text{F}$  capacitor and a load current of  $200\text{mA}$  with 8% ripple full wave rectifier. (2Marks) (6.0 Marks)

2.d. Find the range of load resistance and the maximum power rating for the circuit? (1M +3M +1 M)

(5.0 Marks)

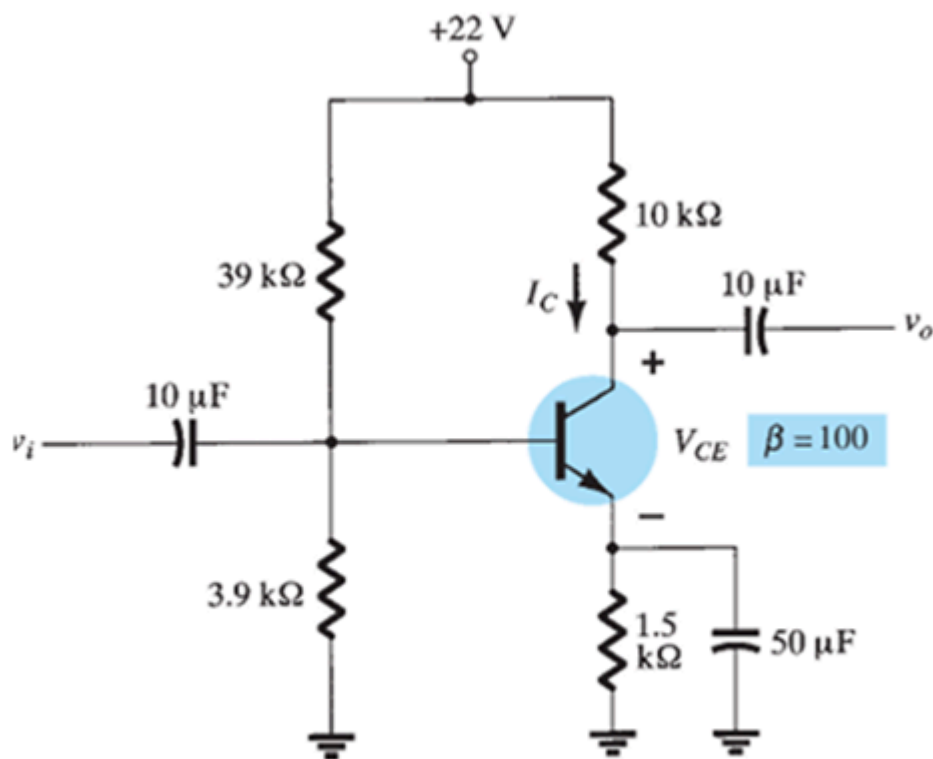


2.e. Compare Bridge Full wave Rectifier with Center-Tap Full wave Rectifier? (4 points) (4.0 Marks)

3.a. Draw the input characteristics of a transistor in CE configuration and briefly explain the input characteristics of the CE configuration. (2M + 3M)  
(5.0 Marks)

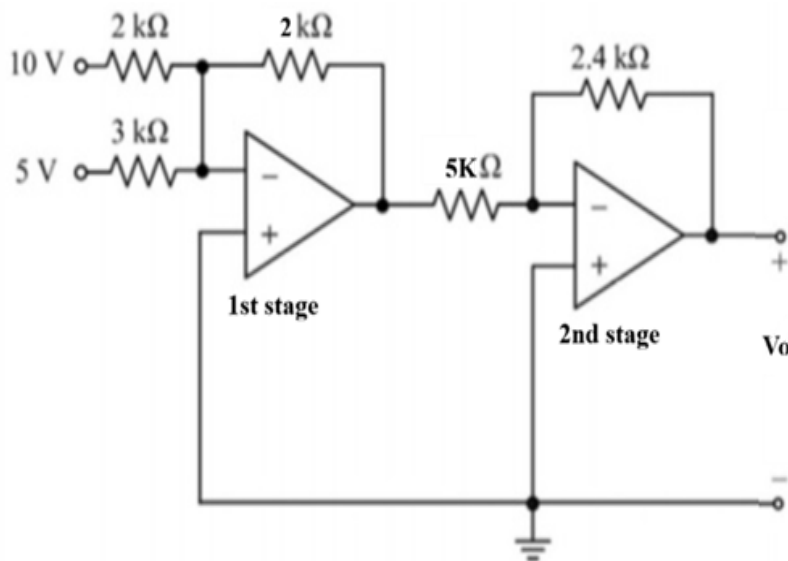
3.b. Determine the dc bias voltage  $V_{CE}$  and the current  $I_C$  for the voltage divider configuration of Fig

(6.0 Marks)



3.c. For the given cascading Op-Amp circuit, answer the following questions: **(2M+1M+2M)**

1. What type of Op-Amp circuit is the 1<sup>st</sup>, and 2<sup>nd</sup> stage?
2. What is the gain of 2<sup>nd</sup> stage?
3. What is the final output voltage  $V_o$ ?

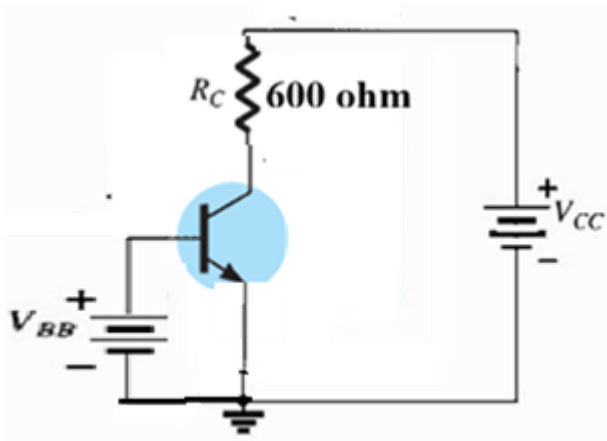


(5.0 Marks)

3.d. With a circuit diagram, derive an expression for the output voltage of op-amp Non-Inverting Amplifier. (5.0 Marks)

3.e. A transistor is connected in CE configuration as shown in the figure, in which the collector supply voltage is 10V, the voltage across the collector resistor is 0.6V. If  $\alpha$  is 0.98 find **(2M+2M)**

1. Collector – Emitter voltage and
2. Base current



(4.0 Marks)

4.a. Draw the circuit diagram, truth table, and timing diagram of 3-bit Asynchronous down counter using T Flip flop. **(2M+2M+2M)** (6.0 Marks)

4.b. 1. Convert the given Boolean function into canonical SOP Form

$$F = A + BC$$

2. Convert the given Boolean function into canonical POS Form

$$F = XY + \bar{X}Z$$

**(3M+3M)**

(6.0 Marks)

4.c.

1. Convert  $[B6A]_H$  to binary.
2. Perform binary addition of  $(29)_{10}$  and  $(-47)_{10}$

**(2M+2M)**

(4.0 Marks)

4.d. Draw the full adder circuit diagram using two half adders. Also write the sum and carry expressions . **(3M+2M)** (5.0 Marks)

4.e. Write the truth table of the 3 inputs XOR gate and draw the circuit of the same using only NAND gates. **(2M+2M)** (4.0 Marks)